

CampusFix AI

Visual Campus Maintenance & Resolution Intelligence System

"Don't fill a complaint form. Show the problem."

DOMAIN

Campus Infrastructure & Student Exp.

HARDWARE

None - Zero Capex, Software Only

AI CORE

Multimodal Vision + Workflow + Verification

STAGE

Hackathon Prototype / Proposed MVP

1 Executive Summary

Every campus continuously generates physical maintenance issues - broken classroom fans, damaged lights, malfunctioning projectors, water leakage, electrical faults, damaged desks and chairs, overflowing waste bins, laboratory equipment problems, and broken doors or windows. These are not occasional events; they are a constant, low-visibility operational load on every institution.

Today, this load is absorbed through verbal complaints, paper forms, phone calls, and generic complaint portals, followed by repeated follow-ups. The result is predictable failure modes: students do not know which department is responsible; maintenance teams receive vague text-only descriptions; multiple students independently report the same problem; administrators lack visual evidence to prioritise work; and students lack visibility into complaint progress. Most critically, traditional systems rarely maintain structured before/after evidence proving that a reported problem was actually addressed.

CampusFix AI proposes a different entry point: instead of asking a student to describe a problem in words, it asks them to photograph it. The system analyzes the image, identifies the affected object and visible problem, classifies category, suggests the responsible department, creates a structured incident, checks for duplicates, and dispatches a task. Once staff complete the repair, they upload an after-repair photo; the vision model compares before/after evidence for AI-assisted verification before authorized staff confirm closure.

CAPTURE -> UNDERSTAND -> CLASSIFY -> ROUTE -> REPAIR -> VERIFY -> CLOSE

"CampusFix AI is not simply another complaint portal. It creates an evidence-based digital lifecycle for physical campus maintenance."

Innovation 1 - Room Scan Mode

One classroom or corridor photograph surfaces multiple candidate maintenance issues at once for student review.

Innovation 2 - Before/After Verification

The system compares original and post-repair photographs to assist - not replace - human verification of reported repairs.

2 The Real Campus Pain Points & Maintenance Gridlock

Campus Maintenance Problem	Current Campus Nightmare	Why Traditional Complaint Systems Fail
Broken classroom equipment (fans, lights, projectors)	Reported verbally or through generic, text-only complaint forms	No visual proof of the fault; staff must inspect blind before triaging
Water leakage / plumbing	Several students may independently report the same physical leak	No de-duplication - one leak becomes many disconnected tickets
Electrical faults	Students may not know whether issue belongs to electrical, maintenance, or IT	Manual triage delays routing; misrouted tickets bounce between departments
Damaged furniture	Low-priority issues remain unresolved with no visibility for weeks	No structured tracking of open vs. resolved status
Housekeeping (bins, sanitation)	Overflowing bins generate repeated, near-identical complaints	No similarity detection to merge repeated reports
Laboratory equipment	Students may not know technical name of equipment or owning department	Free-text descriptions are vague or technically inaccurate
Infrastructure damage (walls, doors, pathways)	Reported without enough visual or locational context	Administrators cannot verify severity remotely

Students typically lack:

- Knowledge of the correct department
- Technical terminology for the issue
- Visibility into complaint status
- Confirmation the issue was actually repaired

Administrators typically lack:

- Centralized visual evidence
- Duplicate detection across reports
- Department-wise workload visibility
- Structured before/after repair verification

3 The Solution: Dual-Engine AI Architecture

"CampusFix AI uses two cooperating engines: a Multimodal Visual Understanding & Incident Intelligence Engine, and an AI-Assisted Maintenance Workflow & Resolution Verification Engine."

PILLAR A - MULTIMODAL VISUAL UNDERSTANDING & INCIDENT INTELLIGENCE

- 1. Visual Problem Detection:** Vision model analyzes affected object, visible problem/damage, likely maintenance category, and supporting context (e.g. ceiling fan -> damaged; tube light -> non-functional). Discloses lighting/angle quality limits.
- 2. Room Scan Mode:** Lab photo surfaces multiple candidate issues in one pass (e.g. Room 204: broken fan, damaged desk, flickering light). Student reviews and confirms before submission; no unconfirmed ticket spam.
- 3. Evidence Extraction (Optional OCR):** Signs, block signage, and asset tags. Absent or illegible text is simply skipped - no hallucinations.
- 4. Duplicate Incident Detection:** Compares open tickets using location, category, semantic similarity, and visual similarity. Flags 'Possible duplicate detected' to enable merging into existing incidents.

PILLAR B - AI-ASSISTED MAINTENANCE WORKFLOW & RESOLUTION VERIFICATION

- 5. Suggested Routing Agents:** Classified category (electrical -> Electrical Maint.; plumbing -> Plumbing; furniture -> Carpentry; IT -> IT Support; cleaning -> Housekeeping). Fully confirmable by staff.
- 6. Maintenance Task Dispatch:** Generates structured task with ID, description, original photo, room location, suggested priority, and assigned staff.
- 7. Before/After Verification Agents (Primary Innovation):** 'after' photo upon repair. Vision model compares both: e.g. 'Resolution evidence appears consistent (96%)'. Staff confirms final closure.
- 8. Resolution Evidence Store:** Stores incident ID, timestamps, assigned department, full audit trail, AI verification score, and human sign-off record.

AI VERIFICATION -> HUMAN CONFIRMATION -> VERIFIED RESOLVED

4 AI Agent & Tool-Calling Specification

Agent / Tool	Trigger	Action	Output
vision.analyze_issue()	Student uploads image/video	Analyzes visible objects & maintenance problems	Candidate object + issue list + confidence
incident.classify_issue()	Vision produces candidate issue	Assigns maintenance category & suggests priority	Category + priority suggestion
incident.detect_duplicate()	New incident submitted	Compares open incidents via location & visual similarity	Duplicate flag + candidate matching incident
routing.assign_department()	Issue category confirmed	Suggests responsible maintenance department	Suggested department record
maintenance.dispatch_task()	Incident confirmed	Creates structured maintenance task with evidence	Maintenance task record & notifications
vision.verify_repair()	Staff uploads after-repair image	Compares before and after visual evidence	AI-assisted resolution verification result
incident.update_status()	Incident changes stage	Transitions: Reported -> Assigned -> In Progress -> Verified	Status updated & audit log persisted
analytics.generate_dashboard()	Dashboard request	Aggregates open, in-progress, pending, resolved counts	Live dashboard analytics data

Human-in-the-loop note: Low-confidence AI decisions involving classification, duplicate detection, routing, or verification are flagged for human review.

5 Competitive Edge & Differentiation

Dimension	Traditional Complaint Portal	CampusFix AI
Reporting	Manual form / verbal complaint	Photograph the problem (1-tap visual capture)
Problem description	Free-text description (vague or inaccurate)	AI-generated structured description & severity
Department assignment	Manual triage & routing delays	AI-assisted, category-based auto-routing
Duplicate complaints	Multiple disconnected tickets for same physical issue	Similarity-based duplicate detection & merge
Visual evidence	Often text-only; inspection blind	Structured before & after visual evidence
Maintenance tracking	Manual follow-up & opaque status	Structured status lifecycle with real-time feedback
Resolution verification	Usually assumed / checkbox checked	AI-assisted image comparison + human confirmation
Dashboard	Limited visibility / siloed spreadsheets	Live incident, workload & resolution dashboard

Traditional: REPORT -> TICKET -> WAIT -> FOLLOW UP

CampusFix AI: SHOW -> UNDERSTAND -> ROUTE -> REPAIR -> VERIFY -> CLOSE

PRIMARY INNOVATION - AI-assisted before/after visual verification

SECONDARY INNOVATION - One classroom photograph can identify multiple candidate maintenance issues

6 Projected Campus Operational Impact

ILLUSTRATIVE PILOT TARGETS / MODELED METRICS - HYPOTHETICAL CAMPUS OF 5,000 STUDENTS, 250 STAFF

Metric	Traditional Process	CampusFix AI	Potential Improvement
Reporting effort	Form fields / free text	Single photograph	Fewer steps per report - 80% faster capture
Duplicate volume	Untracked / repetitive	Similarity-based flagging	Estimated 30%-50% reduction in redundant tickets
Routing time	Manual triage	AI-assisted routing	Immediate hand-off to correct maintenance dept
Follow-up effort	Phone calls / repeated visits	Status dashboard	Reduced inquiries; transparent live tracking
Resolution evidence	Rare / anecdotal	Structured before/after evidence	Consistent audit-grade photo capture
Department visibility	Siloed departmental logs	Unified dashboard	Improved cross-departmental accountability
Department workload	Not centrally tracked	Live aggregation	Optimal resource allocation & staffing insights

Actual operational improvements must be validated through a real campus pilot. Figures represent modeled targets.

7 Presentation & Live Demonstration Blueprint

"CampusFix AI doesn't just collect complaints. It creates evidence that a campus problem was actually solved."

6:00-6:30 min - The Problem: photograph. 'Campuses don't lack technicians; they lack visual connection between problem & workflow.'

0:30-1:00 min - Live AI Demo: Identifies: broken fan, damaged desk, flickering light. User clicks 'Confirm Issues'.

1:00-1:30 min - Automatic Routing: desk -> Maintenance; light -> Electrical. Structured incident tasks created.

1:30-2:00 min - Duplicate Detection: System displays 'Possible duplicate detected - Incident CF-001'. Click 'Merge'.

2:00-2:30 min - Before/After Verification: broken fan. AI: 'Resolution evidence appears consistent (96%)' -> Human confirmation.

2:30-3:00 min - Admin Dashboard: Assigned, In Progress, Awaiting Verification, Verified Resolved, and dept workload.

8 Judge Q&A Defense Sheet

Q: What if the AI incorrectly identifies a problem?

A: CampusFix uses confidence thresholds and human-in-the-loop review. Low-confidence results are flagged for confirmation.

Q: What if students upload irrelevant images / selfies?

A: An upfront vision filter checks campus infrastructure context. Irrelevant or suspicious submissions are rejected immediately.

Q: How does duplicate detection work?

A: Combines location context, issue category, semantic similarity, and visual embedding similarity to detect matching open tickets.

Q: How do you know the repair actually happened?

A: Staff must upload an after-repair photo. Multimodal Vision compares before vs after. Authorized staff make final closure.

Q: Can CampusFix integrate with an existing college ERP?

A: Yes. The architecture exposes REST APIs and webhooks for ERP integration (SAP, Peoplesoft). MVP runs standalone.

Q: Can AI automatically decide urgency?

A: AI suggests priority based on hazard detection (e.g. exposed live wire), but authorized staff confirm critical priorities.

Q: Do you need special hardware?

A: No. Zero Capex. Operates on existing smartphones, tablets, and desktop web browsers.

Q: Is this just a ticketing system?

A: No. Ticketing is only the workflow layer. The AI performs vision understanding, classification, deduplication, and verification.

Q: Do you need to train your own computer vision model?

A: No. The prototype utilizes multimodal LLMs (Gemini Vision) zero-shot. Production can fine-tune on campus imagery.

Q: What if one photo contains multiple problems?

A: Room Scan Mode detects multiple defects simultaneously and lets the user review and selectively log them.

9 Technical Architecture

Layer	Technology	Layer	Technology
Frontend	React + Vite + Tailwind CSS	Authentication	Role-Based Access Control (RBAC)
Backend	FastAPI (Python) or Node.js REST API	Storage	Secure Cloud Object Storage
Database	PostgreSQL / SQLite with vector indexing	Notifications	Web Push Notifications / Email Webhooks
AI Core	Gemini Multimodal Vision API	Integrations	REST Webhooks for Campus ERP

10. Security & Privacy

- Role-based access control (RBAC)
- Secure HTTPS transmission & encryption
- Minimal personal data collection (only room/photo)
- Audit logs for all dispatch & closure actions
- Human-in-the-loop review for all closures

11. MVP vs. Future Scope

- MVP: Photo upload, Room Scan multi-detect
- MVP: Auto-routing, deduplication, admin board
- MVP: Before/after AI verification & sign-off
- Future: WhatsApp chatbot integration
- Future: Voice-based reporting & IoT telemetry

12 Final Project Positioning

Attribute	Specification	Attribute	Specification
Project Name	CampusFix AI	Hardware	None - Zero Capex, Software Only
Full Title	Visual Campus Maintenance & Resolution Intelligence	AI Core	Multimodal Vision + Workflow Automation
Tagline	"Don't fill a complaint form. Show the problem."	Primary Differentiator	AI-assisted before/after visual repair verification
Domain	Campus Infrastructure & Student Experience	Theme	Reimagining the Smart Campus

"CampusFix AI turns a campus from a place where problems are reported into a place where problems become measurable, actionable, and verifiably resolved."